

## Dissociative Disorders

Dissociative disorders are psychiatric disorders in which a person experiences changes in their connection with themselves, their body, emotions, memories, or surroundings. A person may feel as though they are observing themselves from the outside, may perceive their own body or emotions as not belonging to them, or may experience their surroundings as unreal and distant. In some cases, particularly in relation to traumatic experiences, the person may also have difficulty accessing certain memories. Therefore, dissociation can be understood not simply as “forgetting something,” but as a change in the way a person experiences themselves and their experiences as a whole.

Understanding this condition requires consideration of brain regions involved in self-processing, emotion, memory, and cognitive control. Functional neuroimaging studies have demonstrated differences in activation in certain brain regions in individuals with dissociative disorders compared with healthy individuals. These changes have been particularly observed in regions such as the prefrontal cortex, anterior cingulate cortex, amygdala, hippocampus, parahippocampal gyrus, insula, and thalamus. However, these findings do not indicate that dissociative disorders result from a problem in a single brain region. Rather, multiple interconnected brain regions appear to function differently.

This becomes particularly evident in individuals experiencing depersonalization. A person may recognize their own face and know that the person they see in the mirror is themselves, while nevertheless being unable to feel that the face or the self emotionally belongs to them. In studies comparing the processing of one's own face with that of unfamiliar faces, increased activation has been observed in regions such as the right anterior cingulate cortex and medial prefrontal cortex. These findings suggest that there may be a discrepancy between recognizing oneself and experiencing a sense of ownership or familiarity toward oneself. In other words, the problem may not be a loss of the knowledge that “This is me,” but rather a change in the feeling that “This truly belongs to me.”

Differences can also be observed in the processing of emotions. Individuals with depersonalization have demonstrated reduced activation of the amygdala and other limbic regions in response to emotional facial expressions. Since the limbic system plays an important role in the processing of emotional experiences, this finding may be associated with experiencing emotions as less intense or as less personally connected to oneself. Thus, dissociation is not limited to alterations in the sense of self; it is also associated with the way emotions are experienced.

In Dissociative Identity Disorder (DID), differences in brain activity appear to show a more complex pattern. Research has identified different patterns of brain activation between a person's trauma-related identity state and apparently normal identity state. When traumatic memories are re-experienced, changes have been observed in regions such as the amygdala, insula, parahippocampal gyrus, and caudate. In contrast, some trauma-related states have been associated with reduced activation in the prefrontal cortex and regions involved in working memory. In neutral states, increased activation may be observed in certain prefrontal and parietal regions. Therefore, it would not be accurate to say that the brain in dissociative disorders is generally “less active”; rather, the brain appears to function differently across different states.

An important part of these findings points to the relationship between trauma, memory, and emotional regulation. For example, the hippocampus and parahippocampal regions are associated with memory processes, the amygdala with the processing of emotional and threatening information, and the insula with the awareness of internal bodily sensations and emotional states. Differences in the activation of these systems may help explain why traumatic experiences can sometimes be intensely re-

experienced, while at other times they may feel distant, disconnected, or inaccessible.

In conclusion, it is not sufficient to characterize dissociative disorders simply as “a disconnection from reality” or “memory loss.”

Neuroimaging findings indicate that these disorders may involve changes in the connection between the self, body, emotions, and memories. Alterations observed in prefrontal and limbic regions, as well as in areas associated with memory and bodily awareness, may help us understand the neural correlates of dissociative experiences. However, because most studies have involved small sample sizes and have used different methodologies, these findings should not be interpreted as definitive cause-and-effect relationships. Nevertheless, the overall picture is clear: in dissociative disorders, a person does not merely forget something or become disconnected from their surroundings; rather, the way they experience themselves, their emotions, their body, and their past experiences as an integrated whole may change.

## References:

- 1) Purcell JB, Brand B, Browne HA, Chefetz RA, Shanahan M, Bair ZA, Baranowski KA, Davis V, Mangones P, Modell RL, Palermo CA, Robertson EC, Robinson MA, Ward L, Winternitz S, Kaufman ML, Lebois LAM. Treatment of dissociative identity disorder: leveraging neurobiology to optimize success. *Expert Rev Neurother*. 2024 Mar;24(3):273-289. doi: 10.1080/14737175.2024.2316153. Epub 2024 Feb 15. PMID: 38357897; PMCID: PMC10950423.
- 2) Modesti MN, Rapisarda L, Capriotti G, Del Casale A. Functional Neuroimaging in Dissociative Disorders: A Systematic Review. *J Pers Med*. 2022 Aug 29;12(9):1405. doi: 10.3390/jpm12091405. PMID: 36143190; PMCID: PMC9502311

Ali Bocakkıtaç/ Founder, Wionpav Labs/ Medical Student, Faculty of Medicine, Selçuk University